

MSE 2202 — Introduction to Mechatronic Design

Term Project 2026

1 Problem Description

Robotic systems are being developed to collect objects of interest from environments that are presently too dangerous for humans. The primary objective of these robots is to traverse unknown terrains and collect objects of high commercial value. To minimize transportation costs and maximize efficiency, the robots must be compact and lightweight. Further, due the challenges of maintaining reliable communications over long distances, these systems must be able to operate autonomously in a safe manner.

Assuming deployment near an area of interest, each team is required to design and construct an autonomous mechatronic system (a.k.a., a “scavenger” robot) that can depart from a base location, find and gather valuable objects from the ground, return to the base location, and deposit only the retrieved valuable objects into a collection container. Objects are deemed to be valuable based on their reflectance/absorption of specific wavelengths (i.e., their colour). Valuable objects may be identified and/or sorted by the collection robot prior to returning to the base station or by a separate stationary sorting system that is located at the base station. To assist in localizing the collection container or sorting system, a beacon that emits a modulated infrared signal may be used. The entire collection and sorting task must be completed in 120 seconds.

It can be assumed that the area of interest is up to 15 m² in size, is relatively flat and does not have any major obstacles, except at the periphery (outside of the area of interest). The robot should be able to operate within this area without painted/taped lines, markers, wires, or any other installed guides.

It is expected that, once started, the robot will operate fully autonomously at all times. The operation of the robot should be controlled—at no time should it “crash” into anything. Control must be implemented using an ESP32-based microcontroller. Devices must be self powered (i.e., no tethers to stationary power supplies or controllers) and all parts must be brought back to the base station once the task is complete. The overall dimensions of the designed system must allow it to fit within one of the MSE lockers without disassembly. In addition, the use of the motors, drives, and sensors included in the MSE 2202 Lab Kit is encouraged.

The following items will be left to the discretion of each group:

- Method of locomotion
- Type and robustness of navigation and collision avoidance
- Method of collecting objects from the ground
- Method and location of sorting (onboard robot or separate system at base station)
- Method of placing high-value objects into collection vessel

To take advantage of concurrent engineering practice, each student will work in a team of three or four members. Note that the customer needs and primary design specifications are outlined in this document; however, additional design constraints such as durability, ergonomics, simplicity of the user interface, ease of battery charging/replacement, etc. should be identified and considered by the design team. It is expected that each team will utilize accepted product design principles and techniques to progress along the following steps:

- Establish relevant design specifications
- Develop a project plan
- Generate and evaluate several *feasible* design concepts
- Develop the detailed design with CAD
- Develop manufacturing-ready design documentation
- Use design simulation to analyze product performance
- Develop a prototype of design
- Evaluate the prototype performance

The design must be evaluated in terms of product function, design for manufacturing and/or assembly, failure modes, and other relevant performance criteria. The design team must be able to provide sufficient and relevant documentation to justify all critical design decisions.

2 Deliverables

There will be a number of deliverables for this project:

1. A product development file (PDF) documenting the design process
2. A report describing and justifying your design
3. A prototype of your design

2.1 Product Development File

Each group is required to create and maintain a Product Development File (PDF). This is a collection of documents that cover the entire history of the design. In practice, this includes *all of the documents that are generated throughout the design process*, which may include: product design specification lists, Gantt charts, functional decomposition diagrams, quality function deployment charts, system models, sketches of concepts, calculations, go/no-go screening, decision matrices, detail and assembly drawings, circuit drawings, program flowcharts and code, bill of materials, prototype construction notes, failure modes and effects analysis (FMEA), analytical and physical modelling, operating instructions, description of experimental tests, test results and analysis.

The material should be organized according to the different phases of the design process:

1. Specification and Planning Phase
 - (a) Statement of Need/Problem Statement

- (b) Product Design Requirements (understanding the problem)
- (c) Program Plan (planning and scheduling)
- 2. Conceptual Design Phase
 - (d) Conceptual Design Development (concept generation)
 - (e) Concept Evaluation (concept selection)
- 3. Product (Detail) Design Phase
 - (f) Product Generation (refinement of design concepts)
 - (g) Product Evaluation (product design vs. requirements)

The documents and artifacts that comprise the PDF are to be added on an ongoing basis. The PDF will be checked by the teaching assistants during the lab period each week and assigned a score. The assigned scores will be used to adjust the final grade assessed for the complete PDF at the end of the term. Note that the information contained in the PDF is complementary to, and provides the basis for, the product design report.

2.2 Product Design Report

A technical report is to be submitted at the end of the term. The report must clearly describe a single refined solution for the design problem. Proper specifications and well-formatted figures must be included. A typical report will be about 20 pages of double-spaced 11 or 12 pt text. The report length should not exceed 30 pages. General *guidelines* for the content of the report are provided below.

Cover/Title Page:	Outside jacket of the report—contains the name of the course, title of project, and identifying information.
Abstract/Executive Summary:	Abridged version of the entire report written in nontechnical terms; short and informative; normally describes salient features of the report, draws a main conclusion and makes a recommendation; should be written last, after the remainder of the report has been drafted.
Table of Contents:	Shows the content and arrangement of the report; includes a list of appendices, a list of figures, and a list of tables; sometimes followed by nomenclature (set of terms or symbols).
Introduction:	Identifies the design problem and indicates the scope of the report; summarizes key design specifications; provides background information.
Discussion:	Provides all of the details, evidence, and data needed to understand how the product was developed and evaluated; includes a brief description on alternative concepts and why they were rejected; may also describe the adjustments made the prototype phase and the impact of these on the device performance.
Conclusions:	Concise statement of the outcome; evidence that the design objectives have been met.

Recommendations:	If the discussion and conclusions suggest that a specific action needs to be taken, the recommendations state categorically what must be done.
References:	A list of references which were used to conduct the project and which the author considers useful for the reader; follows accepted stylistic conventions and contains sufficient information for the reader to obtain the documents. IEEE reference and citation format is preferred; however, any consistent reference format may be used.
Appendices (Optional):	Supporting data, which rightly belong in the Discussion, but if included, would disrupt and clutter the narrative; inserted in the order that they are referenced in report.

2.3 Prototype

Each group must produce a functional prototype of their design. The prototype should approximate the final design as closely as possible, but need not be exact. It is expected that certain compromises, modifications, and adjustments will be required such that the prototypes can be constructed with the resources available to each team. Remember that each group must design a *product*, not a *prototype*. The following resources may be used for prototype construction:

1. Arduino-based microcontroller—It is expected that the MSEduino boards constructed during Lab 1 will be used; however, *approved* Arduino-compatible ESP32-based microcontroller boards may be considered. The cost of an alternate board must be included in the overall cost (see additional components below).
2. MSE 2202 Lab Kit—Any of the components included in the lab kit are suitable for inclusion. The use of these components is encouraged.
3. Electronic components—Electronic components in addition to those available in the MSE 2202 lab kits may be obtained through the Electronics Shop. Each team will be provided with a budget of \$20. These components must be returned at the end of the term.
4. Rechargeable battery packs—The Power Bank battery included in the MSE 2202 lab kit **must be used in an unmodified form**. It is expected that the device(s) will operate within the energy budget permitted by one of these batteries.
5. Rapid prototyping machine—Two fused-deposition modelling (FDM) systems that can construct parts from layers of extruded PLA or ABS plastic are available in the Mechatronics Lab. Build costs are \$0.08/g of material. Note that, the turnaround time on component builds (especially for large parts) may be up to one week. All build requests are at the discretion of Eugen Porter.
6. Laser cutter—A 60 W CO₂ laser is available in the Mechatronics Lab. It is capable of cutting 2D profiles from a variety of sheet goods including cardboard, acrylic, plywood, pressboard, and corrugated plastic up to 1/4" thick. For safety reasons, only approved materials will be cut. As with the rapid prototyping machines, the turnaround time to have parts fabricated may be up to one week. The combined budget for rapid prototyping and laser cutting is \$20.

7. Additional components purchased or salvaged—Each group can add any additional components required by their design. The total cost of all additional components cannot exceed \$50. To ensure fairness, receipts are required; for salvaged components, the fair-market value, as determined by advertised prices in catalogs or on websites, must be provided.

3 Milestones

Four project milestones are required to be submitted electronically on OWL during the term. These are as follows:

1. Concept Generation and Selection, due Friday, **February 13, 2026**. This report should include concepts generated for individual subsystems, along with the overall design of the robot. The most promising overall design should be selected and justified.
2. Detail Design documentation, due Friday, **March 6, 2026**. This includes a complete CAD model (eDrawing + SolidWorks) and the bill of materials for the robot.
3. Drive System implementation, due Monday, **March 16, 2026**. A brief video that demonstrates the fully functional robot drive system, along with corresponding microcontroller code.
4. Pickup and Sorting System implementation, due Monday, **March 23, 2026**. A brief video that demonstrates the fully functional pickup and sorting system(s), along with corresponding microcontroller code. In addition, the procedure used to determine appropriate threshold(s) for the colour sensor must be provided.

The milestones are worth 5%, 4%, 3%, and 4% the overall course grade, respectively. Failure to submit a milestone will result in a project grade of 0.

4 Prototype and Demonstration

Each team is required to submit a digital video to the instructor showing the capabilities of their fully-functional design by 11:55 pm on Monday, **March 30, 2026**. The video should demonstrate the efficacy of their design with respect to each of the tasks outlined in the problem description and clearly show the functionality of the complete system.

A showcase of the robots developed by each group will be held on Tuesday, **March 31, 2026** from 8:30–10:30 am. The showcase will provide each group with an opportunity to demonstrate their robot system to the rest of the class. Points will be awarded to each team according to the following formula:

$$\text{Score} = \frac{(200 - t) \max((g - 2b), 0)^2}{w}$$

where t = the time in seconds taken to collect and sort the valuable objects, g = the number of high-value objects in the collection container, b = the number of other objects in the collection container, and w = the combined mass of the robot and sorter in kg. In order to count, the system must complete the task within the 120 second (2 minute) time limit. Collected objects that are not deposited into the collection container will not be scored.

Following the showcase, the overall mechanical and electrical build quality of the prototype will also be assessed. The prototype must be placed in the team locker in ACEB 3445 on, or before, Thursday, April 9, 2026.

5 Mark Distribution

To make the grading process fair, each member of the design team will evaluate every member of the group (including themselves) for the percent of their contribution to the design project. These confidential evaluations must be justified and supported by written commentary. The evaluations will be averaged to find the contribution and individual grade for each student.

For example, if there are four students in a group and each student makes an equal (25%) contribution then all members will receive the same grade for the project. However, if the individual contributions vary, then the grades will be corrected by the difference from 25%. For example, if the group mark for the project was 80% and student A makes a 40% contribution to the project then his/her mark will be 95% (i.e., $80 + (40 - 25)$). Furthermore, if student B in the same group makes a 30% contribution then that student will be assigned a mark of 85% (i.e., $80 + (30 - 25)$). Finally, if students C and D make contributions of 20% and 10% then their individual marks will be 75% and 65%, respectively. Note that the maximum possible mark for the project is 100%.

6 Project Deadline

The PDF, product design report, and prototype are due no later than 5 pm on **Thursday, April 9**. *Late submissions will not be accepted.* Marks will be assigned on the basis of innovation, correctness of solution, thoroughness of analysis, clarity of presentation, and neatness.

Note: The above design requirements and deliverables are subject to adjustments and changes as necessary.